



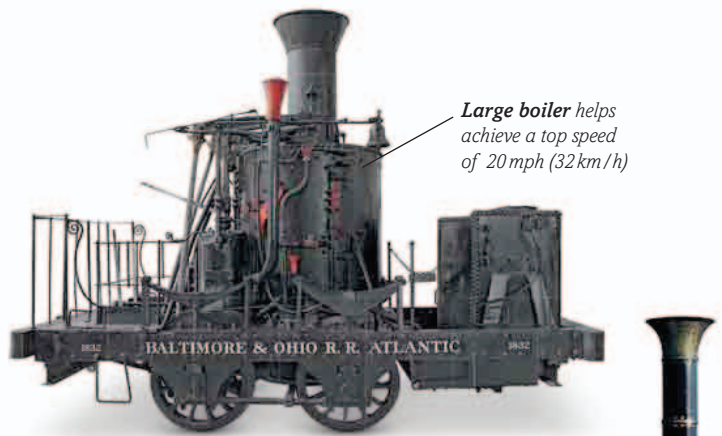
Tubes inside boiler are made of gun barrels

Tom Thumb
USA 1830

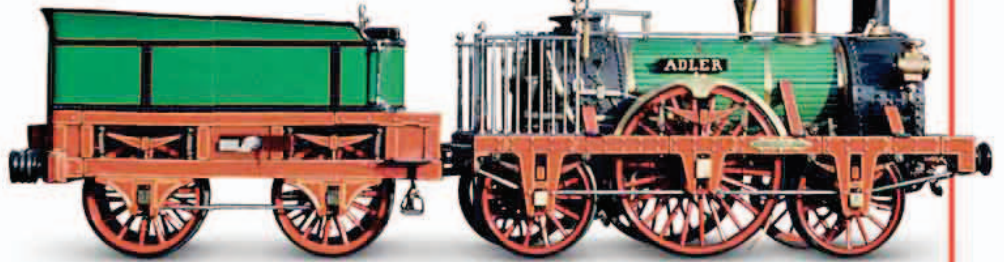


Large boiler helps achieve a top speed of 20 mph (32 km/h)

B&O Atlantic USA 1832



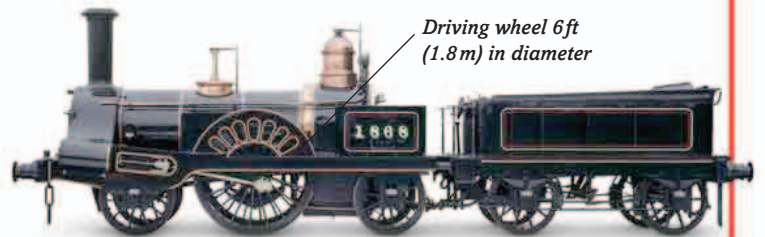
Carriages pulled by horses ran on the same lines as the **Adler**.



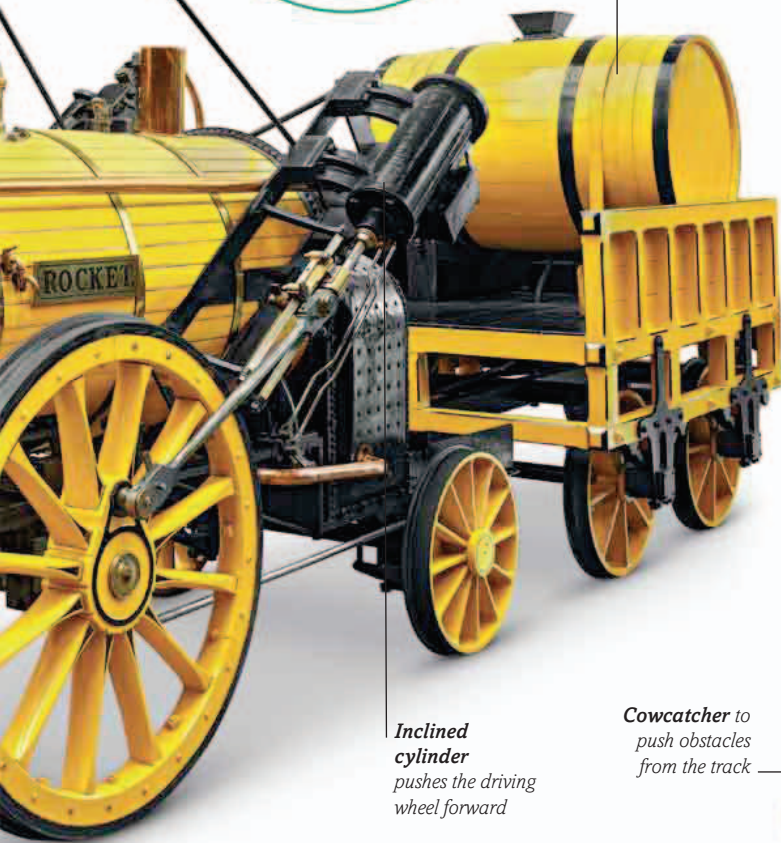
Adler UK/Germany 1835

Water barrel

Driving wheel 6ft (1.8m) in diameter



GJR Columbine UK 1845

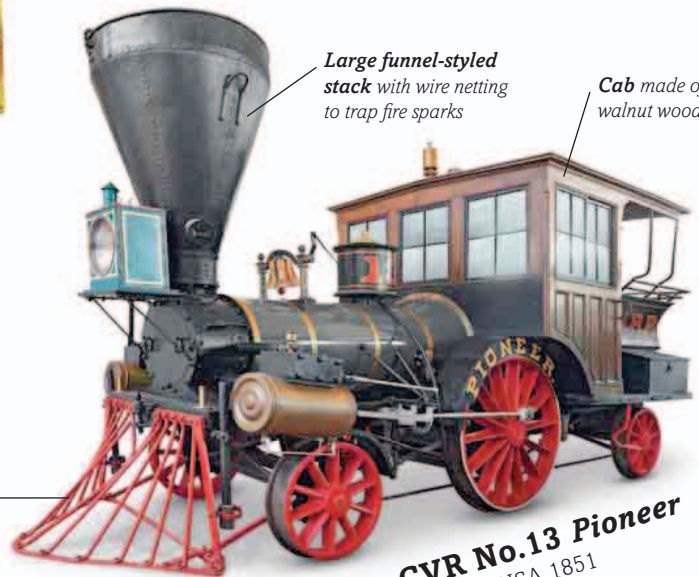


Inclined cylinder pushes the driving wheel forward

Cowcatcher to push obstacles from the track

Large funnel-styled stack with wire netting to trap fire sparks

Cab made of walnut wood



CVR No.13 Pioneer
USA 1851

In 1829, Robert Stephenson's **Rocket** beat the **Novelty** at the Rainhill Trials in the UK, where engines competed to run on the Liverpool and Manchester Railway—the world's first intercity line. Stephenson's company later built the **Adler**, the first German commercial train. John Steven's

Steam Wagon was the first American train, but it ran on a small circular track. The first engine used on regular service in the USA was **Tom Thumb** on the Baltimore & Ohio Railroad (B&O). By 1840, the country had over 2,796 miles (4,500 km) of track, more than found in the whole of Europe.